M378K Introduction to Mathematical Statistics
Problem Set #17
Relative efficiency.

Definition 17.1. *Given two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$, the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is defined as*

$$\boxed{\mathit{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\mathrm{Var}[\hat{\theta}_2]}{\mathrm{Var}[\hat{\theta}_1]}.}$$

Problem 17.1. *Let Y_1, Y_2 be a random sample from the exponential distribution with the unknown parameter θ.*

(i) *The estimator $\hat{\theta}_1 = (Y_1 + Y_2)/2$ for θ is proposed. What is its variance?*

(ii) *The estimator $\hat{\theta}_2 = cY_{(1)}$ for θ is proposed. Find the constant c such that $\hat{\theta}_2$ is an unbiased estimator of θ. What is its variance?*

(iii) *Calculate the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$.*

independent

i. $\mathrm{Var}[\hat{\theta}_1] = \mathrm{Var}\left[\frac{1}{2}(Y_1+Y_2)\right] = \frac{1}{4}\mathrm{Var}[Y_1+Y_2] = \frac{1}{4}(\mathrm{Var}[Y_1]+\mathrm{Var}[Y_2])$

$= \frac{\mathrm{Var}[Y_1]}{2} = \frac{\theta^2}{2}$

ii. $Y_{(1)} \sim \cancel{?}\ E\left(\frac{\theta}{2}\right)$

$\mathbb{E}[\hat{\theta}_2] = \theta \Rightarrow \boxed{c=2}$

$\mathrm{Var}[\hat{\theta}_2] = \mathrm{Var}[2\cdot Y_{(1)}] = 4\cdot\mathrm{Var}[Y_{(1)}] = \cancel{4}\cdot\frac{\theta^2}{\cancel{4}} = \theta^2$

iii. $\mathrm{eff}(\hat{\theta}_1, \hat{\theta}_2) = \dfrac{\theta^2}{\frac{\theta^2}{2}} = 2$

$\square$

M378K Introduction to Mathematical Statistics
Problem Set #18
Consistency.

Definition 18.1. *$\hat{\theta}_n$ is said to be a* consistent estimator *of θ if*

$$\hat{\theta}_n \to \theta \quad \text{in probability as } n \to \infty,$$

i.e., if for any $\varepsilon > 0$,

$\mathbb{E}[\hat{\theta}_n]$

$$\lim_{n\to\infty} \mathbb{P}\left[|\hat{\theta}_n - \theta| > \varepsilon\right] = 0.$$

Theorem 18.2. *Let $\hat{\theta}_n$ be* unbiased *and such that*

$$\mathrm{Var}\left[\hat{\theta}_n\right] \xrightarrow{n\to\infty} 0.$$

Then, $\hat{\theta}_n$ is a **consistent estimator**.

Markov Inequality
Chebyshev Inequality

Problem 18.1. *Let $Y_1, Y_2, \ldots, Y_n$ be a random sample from any distribution with finite first and second moments. Propose a consistent estimator for the population mean μ and* **prove** *that it is, indeed, consistent.*

→: Propose: $\bar{Y}_n = \frac{1}{n}(Y_1 + \cdots + Y_n)$

Criteria for consistency: ✓

- unbiased ✗ ✓
- $\underbrace{\mathrm{Var}[\bar{Y}_n]}_{} \xrightarrow[n\to\infty]{?} 0$

$$= \frac{\mathrm{Var}[Y_1]}{n} \xrightarrow[n]{?} 0$$

$$\frac{\overset{<\infty}{\mathbb{E}[Y_1^2]} - \overset{<\infty}{(\mathbb{E}[Y_1])^2}}{n} \xrightarrow[n]{} 0$$

Google: Cauchy Dist'n.

Problem 18.2. *Consider a random sample $Y_1, Y_2, \ldots, Y_n$ from a power distribution with the density of the form*

$$f_Y(y) = \theta y^{\theta-1} \mathbf{1}_{(0,1)}(y).$$

*What is a consistent estimator for $\frac{\theta}{\theta+1}$? **Prove** that your choice is indeed consistent.*

$\rightarrow$: $\mathbb{E}[Y_1] = \int_0^1 y \cdot \theta \cdot y^{\theta-1} dy = \theta \int_0^1 y^{\theta} dy = \theta \cdot \left(\frac{y^{\theta+1}}{\theta+1}\right)_{y=0}^{1}$

$$\boxed{\mathbb{E}[Y_1] = \frac{\theta}{\theta+1}}$$

We propose: $\bar{Y}_n$

- unbiased ✗ ✓
- $\mathrm{Var}[\bar{Y}_n] \xrightarrow[n\to\infty]{?} 0$

$\mathrm{Var}[\bar{Y}_n] = \frac{\mathrm{Var}[Y_1]}{n}$

If $\boxed{\mathrm{Var}[Y_1]} < \infty$, we're done ☺

$\boxed{\mathrm{Var}[Y_1]} = \boxed{\mathbb{E}[Y_1^2]} - (\mathbb{E}[Y_1])^2 < \infty$

$\boxed{\mathbb{E}[Y_1^2]} = \int_0^1 y^2 \cdot \theta \cdot y^{\theta-1} dy = \theta \cdot \int_0^1 y^{\theta+1} dy = \frac{\theta}{\theta+2} < \infty$

$\square$

Example. • $Y_1, \dots, Y_n$ random sample from $E(\tau = \theta)$

$\hat{\theta}_n = n \cdot Y_{(1)}$ unbiased for all n

$\sim E\left(\frac{\tau}{n}\right)$

Consider $Var[\hat{\theta}_n] = Var[n \cdot Y_{(1)}] =$

$= n^2 \cdot Var[Y_{(1)}] = n^2 \cdot \frac{\theta^2}{n^2} = \theta^2$

$\sim E\left(\frac{\tau}{n}\right)$

$\not\to 0$

The criterion is not satisfied.

• Task: Find an unbiased estimator for θ in

$Y_1, \dots, Y_n \sim U(0, \theta)$

Is it consistent?